

Networking Summary

- ❖ **Network:** consist of two or more computers that're linked in order to share resources, exchange files or allow electronic communications.
- ❖ **Purposes of the network**
 - ✚ To share resources such as files
 - ✚ To communicate / send information
 - ✚ To ease administration /SI uu u fududeeyo maamulka
 - ✚ To protect and backup the data
- ❖ Two PC's can connect together using **UTP** crossover
- ❖ Five computers can connect together using **HUP** to share information
- ❖ Two switch can connect together using **Router** to share data.
- ❖ Requirements for creating network
 - ✚ More than one computer/device
 - ✚ An Operating System (OS)
 - ✚ Network Interface Card (NIC)
 - ✚ Protocol (IP/TCP)
 - ✚ Transmission media (Guided and Unguided)
 - ✚ Switch, Hup and Router
- ❖ Types of Network
 - ✚ **Personal Area Network (PAN):** smallest type of network and has limited span, interconnect personal devices. such as those that are Bluetooth enabled such as Smartphones, headphones, mouse, printers and so on.
 - ✚ **Local Area Network (LAN):** is a network commonly used to interact and share computer resources inside a building or multiple buildings in limited area.
 - ✚ **Metropolitan Area Network (MAN):** spans multiple buildings in a city area.

✚ **Wide Area Network (WAN):** much larger than other network types and can span many areas, such as cities, states or countries.

❖ **Network Interface Card (NIC):** hardware component typically circuit board or chip installed on computer so it can connect to a network.

NOTE: NIC is the computer's entry and exit point

✚ Functions of NIC

- Physical connection to the network
- Data transmission and Reception

❖ Definition of Local Area Network in terms of **Protocol** and **Topology**

✚ In terms of **Protocol:** is a set of rules established (setup or created) for the users to gain control of the network to exchange information.

✚ In terms of **Topology:** is the network architecture used to interconnect the networking equipment

❖ **Point-to-point topology:** is a type of topology that works with the functionality of the sender and receiver.

✚ It the simplest communication between two nodes which one is the sender and the other one is receiver.

✚ Provides high bandwidth.

❖ **Ring topology:** is a specific type of LAN topology that connects each device to exactly two of its neighboring devices for communication purposes.

NOTE: if any cable or device in the network fails, it will break the loop and cause entire segment to go down

❖ **IEEE:** stands for Institute of Electrical and Electronics Engineers

❖ **Bus Topology:** is a network type in which every computer and network device is connected to a single cable.

✚ It's bi-directional

- ✚ It's multi-point connection
- ✚ Non-robust topology
- ❖ **Star topology:** all devices are connected to a single hub through a cable.
 - ✚ the hub can be intelligent known as an active hub
 - ✚ Features of star topology
 - All networking devices connect to central hub or switch.
 - The networking devices do not share data connections to/from the central hub or switch.
 - A hub broadcasts all data traffic to all networking devices connected to its data ports.
- ❖ **Mesh topology:** every device is connected to another device via particular channel.
- ❖ Protocols used in mesh topology are:
 - ✚ **AHCP** – Ad Host Configuration Protocol
 - ✚ **DHCP** – Dynamic Host Configuration Protocol
- ❖ Features of mesh topology
 - ✚ All networking devices are heavily interconnected.
 - ✚ Unnecessary/excessive data traffic paths
 - ✚ Expensive
- ❖ **OSI** – Open Systems Interconnection was developed in 1984 to enable different types of networks to be linked together.
- ❖ The intent of OSI model is to provide framework for networking that ensures compatibility in the network hardware and software, and to accelerate the development of new networking technologies.
- ❖ Advantages of OSI Reference Model
 - ✚ Simplifies communication by dividing it into layers for easier development, design, and troubleshooting.
 - ✚ Allows products from different vendors to work together.

- ✚ It ensures that changes made in one layer do not affect other layers.
- ✚ It enables different hardware and software to communicate easily
- ❖ **Layer 7:** responsible for providing an interface for the user to interact with applications or network services
- ❖ **Layer 6:** defines specific format for the data to be sent over the network.
- ❖ **Layer 5:** provides the control functions necessary to establish, manage and terminate the connections as required to satisfy the user request
- ❖ **Layer 4:** responsible for end-to-end transportation between hosts. e.g of protocol are **TCP** and **UDP**
- ❖ **Layer 3:** provides connectivity and path selection between two nodes. e.g are **IP** and **IPX**
- ❖ **Layer 2:** defines how data is formatted for transmission and how access to the network is controlled.
 - ✚ Its considered "*Media Access Control Layer*" and it's where MAC address is defined
 - ✚ **MAC Address:** is a hardware address and is 12 digit Hexadecimal identifier used to point out the devices in the network. e.g are **Token ring** and **Ethernet**
- ❖ **Layer 1:** it transfers data into bits format.
 - ✚ Conversions of data depends on the type of media used:
 - **Copper Media:** an electrical signal of different voltages.
 - **Fiber Media:** light pulses of different wavelengths.
 - **Wireless Media:** radio frequencies

Layer	Function	Examples
7. Application	Support for applications	-HTTP, FTP, SMTP (email)
6. Presentation	Protocol conversion, data translation	ASCII, JPEG
5. Session	Establishes, manages, and terminates sessions	NFS, SQL
4. Transport	Ensures error-free packets	TCP, UDP
3. Network	Provides routing decisions	IP, IPX
2. Data link	Provides for the flow of data	MAC addresses
1. Physical	Signals and media	NICs, twisted-pair cable, fiber

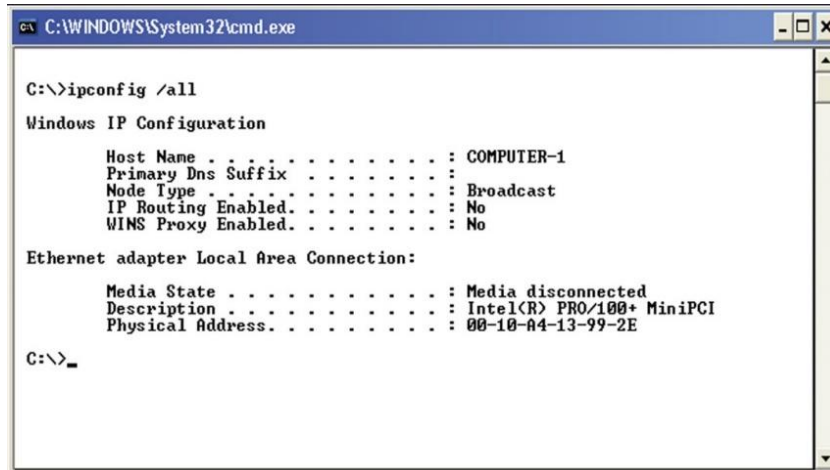
- ❖ the network administrator uses OSI model for troubleshooting network problems by verifying functionality of each layer.
- ❖ Assume that a network is having problems accessing an email server (**a layer 7 application**).
- ❖ The first troubleshooting step for the network administrator is to ping the IP address of the email server (**layer 3 test**).
- ❖ This could be due to a cabling problem (**layer 1**) or a problem with a switch (**layer 2**), or a router (**layer 3**), or the server could be completely down (**layer 7**).
- ❖ The networking administrator needs to have a good understanding of the all seven layers of the OSI model.
- ❖ There are three basic steps in the process of isolating the network problem.

- ✚ Is the connection to the machine down? (**layer 1**)
 - ✚ Is the network down? (**layer 3**)
 - ✚ Is a service on a specific machine down? (**layer 7**)
-
- ❖ Ethernet is a technology commonly used in wired LAN
 - ❖ **A LAN** is a network of electronic devices that covers a small area such as office or building
 - ❖ **Ethernet:** is a communication protocol that governs the transmission of data over a LAN and is commonly known as **IEEE 802.3 protocol**.
 - ❖ The networking protocol used in most modern computer networks is **Ethernet**.
 - ✚ Ethernet is a CSMA/CD protocol.
 - **CS:** Carrier Sense
 - **MA:** Multiple Access
 - **CD:** Collision Detection
 - ❖ The information in an Ethernet network is exchanged in a packet form
 - ❖ The packet provides grouping of the information for transmission.

Chapter 1 part 2 Addresses

- ❖ **A Network Address:** is a logical or physical address that uniquely identifies a host or a machine in a telecommunication network.
- ❖ Examples of network addresses are
 - ✚ IP addresses
 - ✚ MAC addresses
 - ✚ Telephone numbers
- ❖ Types of addresses are
 - ✚ **Physical or Media Access Controller (MAC)** addresses
 - ✚ **Logical or Internet Protocol (IP)** addresses

- ❖ MAC address is also called **Ethernet Address, Physical Address, Hardware Address or Adapter Address**
- ❖ The MAC address can be obtained from computers operating under Microsoft windows by typing **ipconfig/all** command in the CMD



```

C:\WINDOWS\System32\cmd.exe

C:\>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : COMPUTER-1
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Broadcast
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Description . . . . . : Intel(R) PRO/100+ MiniPCI
    Physical Address. . . . . : 00-10-A4-13-99-2E

C:\>

```

- ❖ **MAC address:** is a 12 character unique identifier for electronic devices on a network. An example of MAC is 00-B0-D0-63-C2-26
 - ✚ The first hexa-decimal cods are used to indicate the vendor of the network interface. Also called **Organizationally Unique Identifier (OUI)**
 - ✚ the last 6 hexa-decimal values are unique numbers assigned by vendor.

**Company ID / Vendor
Serial #**

Manufacturer

00-AA-00-B6-7A-57

Intel Corporation (00-AA-00)

00-00-86-15-9E-7A

Megahertz Corporation (00-00-86)

00-50-73-6C-32-11

Cisco Systems, Inc. (00-50-73)

00-0A-27-B7-3E-F8

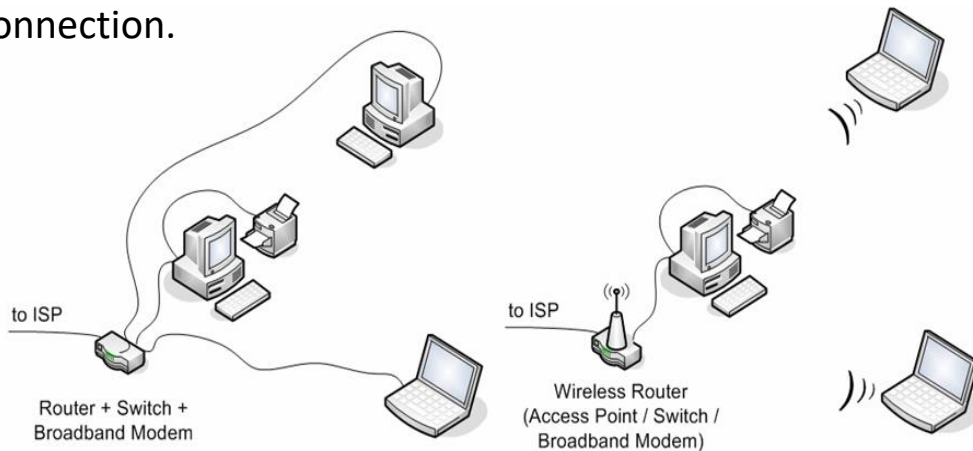
Apple Computer Inc. (00-0A-27)

- ❖ **An Internet Protocol (IP) address:** is unique numerical identifier for every device or network that connects to the internet.
 - ✚ Versions of IP address are **IPv4** and **IPv6**
 - ✚ **IPv4 address** consists of **four dotted decimal numbers** (**192.168.35.4**) the first 3 octet represent particular network on the internet while the rest represent actual host address such as workstation or server.
 - Range for each of the decimal values is 0 to 255
 - ✚ **IPv6 address** represent **eight groups of four hexadecimal digits** separated by colons such as
2620:cc:8000:1c82:544c:cc2e:f2fa:5a9b
- ❖ Aspects of IPv4 address is divided into two
 - ✚ **Network portion**
 - ✚ **Host portion**
- ❖ Types of IPv4 address
 - ✚ **Public IP address:** provided by ISP to access to the internet. It's not free but unique
 - ✚ **Private IP address:** it's set aside for use in private internets or local networks. It's free cost and can be duplicated
NOTE: private address are not valid address for internet usage, it's used within private LAN to create an IP network
- ❖ **IANA** – Internet Assigned Numbers Authority: is the organization that manages and distributes the IP entire the globe.
- ❖ **A Subnet mask:** is a 32-bit number used to separate the network address from the host address in IPv4 Address.

Default Subnet Mask

Class A	N. H.H.H	255. 0.0.0
Class B	N.N. H.H	255.255. 0.0
Class C	N.N.N. H	255.255.255. 0

- ❖ **A wired network:** uses cabling and connectors to establish the network connections.
- ❖ **A wireless network:** uses radio signals to establish the network connection.



- ❖ Advantages of wired network
 - + Faster data transfer
 - + Low setup cost
 - + Less interference
- ❖ Disadvantages of wired network
 - + Specialized tools are needed for cable connection
 - + Cable installation can be time consuming and require effort
- ❖ Advantages of wireless network
 - + User mobility
 - + Simple installations
 - + No cables required
- ❖ Disadvantage of wireless network
 - + Security issues
 - + Slower data transfer compared to wire network
- ❖ Network adapters both wired and wireless are available.
- ❖ The wireless-N adapter is designed to fit into a laptop's PC Card slot, offering faster data transfer than wireless-G



- ❖ A wireless router connects directly to a modem by a cable, this allows it to receive information from – and transmit information to – the internet.



- ❖ **ADSL modem:** is device that connects your home network to the internet via the telephone, enabling broadband services like internet, phone, and TV. The broadband is provided by the phone company or separate ISP.



- ❖ **Access point:** is a device that connects wireless devices to the wired LAN enabling data transfer between them.



- ❖ The Essential requirement needed when planning the home network.
 - ✚ Data speed
 - ✚ Cost
 - ✚ Ease of implementation
 - ✚ Appearance
 - ✚ Home access
 - ✚ Public access
- ❖ To protect your home network follow these steps
 - ✚ Change the default factory password
 - ✚ Change the default SSID
 - **SSID:** is the name used by your router or access point to identify and connect wireless devices to your network.
 - ✚ Turn off the SSID broadcast
 - Hotspots make it easy for the user to gain wireless access.
 - ✚ Turn encryption on
 - To secure your network, turn on encryption, by choosing **WEP** – Wired Equivalent Privacy *or* **WPA** – Wi-fi Protected Access. **WPA2** offers better security and works with device that support WPA
- ❖ **NAT – Network Address Translation:** translates the private IP address to a public address for routing over the internet.
 - ✚ NAT uses **PAT – Port Address Translation** to attach a port number to the connection

Chapter 2 Cabling

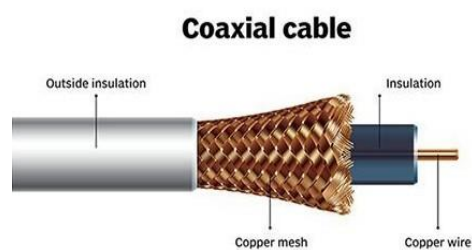
❖ **Media:** is a physical pathway that connects network devices and carries data between components.

❖ **Transmission media:** is a pathway that carries the information from sender to receiver.

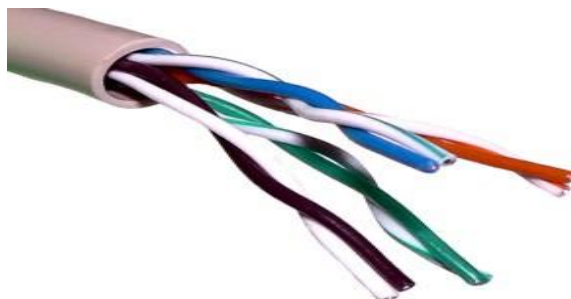
✚ The two main types of transmission media are

- **Guided:** is physical transmission media which can be seen and touched. It has three types
 - **Coaxial cable**
 - **Twisted pair cable**
 - **Fiber optic cable**
- **Unguided:** is media used for data transmission which can't be seen or touched. It has three types
 - **Radio waves**
 - **Micro waves**
 - **Infrared waves**

❖ **Coaxial cables:** are guided media that carry higher frequency signals than twisted pair cable. They have a copper core, plastic jacket and a braided shield.



❖ **A twisted Pair Cable:** a cable made up of numerous independently wrapped wires that are coiled around one another.



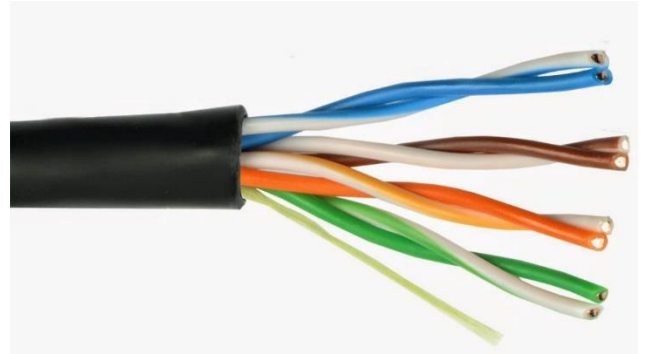
❖ Types of twisted pair cable are

✚ **STP – Shielded Twisted Pair:** is a twisted pair cable enclosed in foil or mesh shield

✚ **UTP – Unshielded Twisted Pair:** is a cable with wire that are twisted together.



Shielded Twisted Pair (STP)



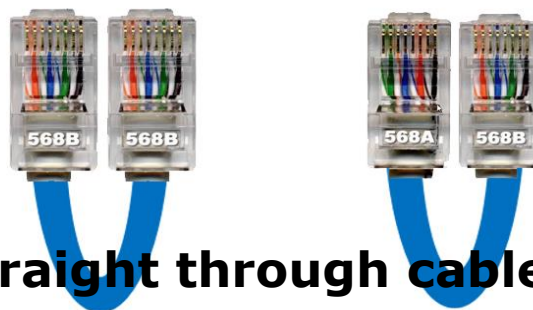
Unshielded Twisted Pair (UTP)

NOTE: UTP cables are lighter and thinner than STP

UTP Cable Categories

Type	No. of Pairs	Transmission Rate	Applications
Cat 1	2 twisted wire pairs	Voice Grade	An old Telephone Cable
Cat 2	4 twisted wire pairs	4 Mbps	Token Ring N/W
Cat 3	4 twisted wire pairs	10 Mbps	Token Ring N/W
Cat 4	4 twisted wire pairs	16 Mbps	Token Ring N/W
Cat5/5e (enhanced)	4 twisted wire pairs	100 Mbps	E, FE or GE
Cat 6/6a (advanced)	4 twisted wire pairs	1000 Mbps with 250-500 MHz B/w	FE or GE
Cat 7	4 twisted wire pairs	10 Gbps with 600 MHz B/w	GE

- ❖ **Cat5** cable was created in 1991
- ❖ Fast Ethernet refers to the 100Mbps data in Ethernet network.
- ❖ **Terminations:** is the point at which the cable connects to terminals in a modular plug, jack or patch panel.
- ❖ Twisted pair cables often called **Patch cables**, are used in networks to connect networking components.
- ❖ **A Straight through cable:** is a twisted pair cable used in LANs to connect a DTE device, like a computer, to a hub or switch which is a DCE device.
 - ✚ **DTE – Data Terminal Equipment**
 - ✚ **DCE – Data Circuit-terminating Equipment**
- ❖ **A crossover Ethernet cable:** is a form of Ethernet connection that connects one DTE device to another or one DCE to another.



Uses of straight through cables

- ✓ Switch to router
- ✓ Switch to PC or server
- ✓ Hub to PC or server
- ✓ Bridge to router
- ✓ Bridge to PC or Server

Uses of Crossover

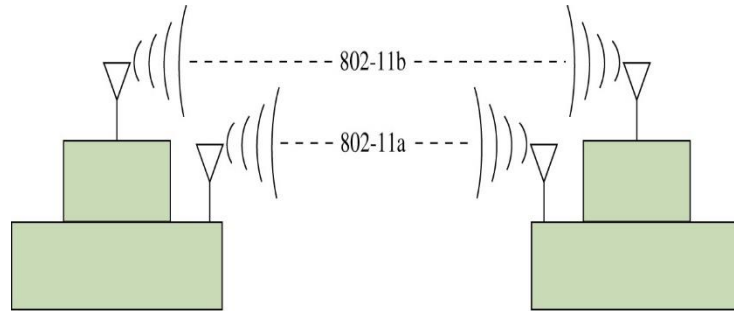
- ✓ Switch to switch
 - ✓ Switch to hub
 - ✓ Hub to hub
 - ✓ Router to router
 - ✓ Router Ethernet port to PC NIC
 - ✓ PC to PC
-
- ❖ The difference between 568A and 568B lies just color arrangement, and there is no performance difference.
 - ❖ **Ethernet:** is a traditional technology for connecting devices in a wired LANs or WANs, allowing communication through a common protocol.
 - ❖ Ethernet copper supports
 - ✚ 10 Mbps (Ethernet)
 - ✚ 100Mbps (FastEthernet)
 - ✚ 1000Mbps (gigabit Ethernet),
 - ✚ 10Gbps (ten gigabit Ethernet – 10G).
 - ❖ Advantages of Ethernet
 - ✚ Low cost
 - ✚ Backward compatibility
 - ✚ Resistance to noise
 - ✚ Good data transfer quality
 - ✚ High speed
 - ✚ Reliability
 - ❖ Disadvantage of Ethernet
 - ✚ Best for smaller, shorter networks
 - ✚ Limited mobility

- + Longer cables may cause crosstalk
 - + Not ideal for real time or interactive applications
 - + Speeds drop with higher traffic
 - + Troubleshooting can be difficult especially when identifying faulty cables or nodes.
- ❖ **Labeling:** is extremely important for all aspects of a computer network and data center. It's types are
- + **Port labeling**
 - + **System labeling**

Chapter 3 Wireless Networking

- ❖ **Wi-Fi – Wireless:** is an interconnection between the network devices to share the resources and data without using any physical media.
- ❖ **The PHY – Physical Layer:** defines the method of transmitting the data. e.g
- ❖ The five physical layer technologies 802.11 are
 - + **RF – Radio Frequency** and
 - + **Infrared**
 - + **DSSS – Direct Sequence Spread Spectrum** - Most often used
 - + **FHSS – Frequency Hopping Spread Spectrum**
 - + **OFDM – Orthogonal Frequency Division Multiplexing**
- ❖ Wireless agencies and standards
 - + **IEEE 802.11:** creates and maintains operations at standards.
 - + **Wi-Fi alliance:** is an organizations whose function is to test and certify wireless equipment for compliances with IEEE 802.11 standards.
 - + **Licensed Frequency:** business companies pay licensing fee for an exclusive right to transmit.
 - + **Unlicensed Frequency:** No need any permission to use this frequency.

- ❖ IEEE 802.11a equipment is not compatible and interfere with 802.11b



- ❖ The 802.11g standard supports the higher data transmission rates of 54 Mbps but operates in the same 2.4 GHz

- + The 802.11b wireless clients will be able to communicate with the 802.11g access point vice-verse.
- + **Multiple Input Multiple Output – MIMO**: splits data into multiple spatial streams , transmitted via separate antennas.
- + **MIMO Power Save Mode**: uses multiple paths only when needed conserving power.
- + **Multiuser MIMO - MUMIMO**

- ❖ **Wi-Fi** stands for **Wireless Fidelity**

- ❖ **MAC – Media Access Control layer** ensures data reliability, control access to the wireless medium and protects data privacy.

- ❖ Wireless management protocol services are:

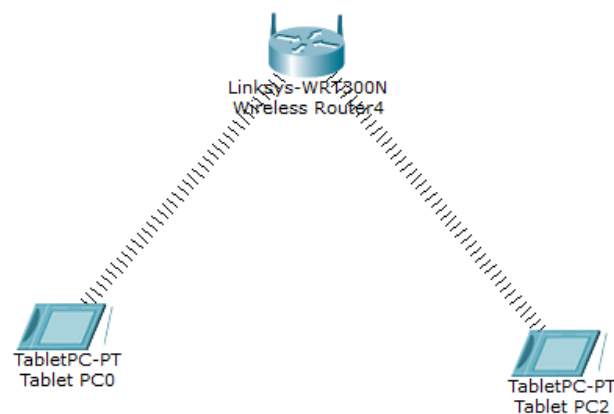
- + Authentication
- + Association
- + Data delivery
- + privacy

- ❖ **WLANs – Wireless LANs**: provide mobility to communicating nodes within coverage area.

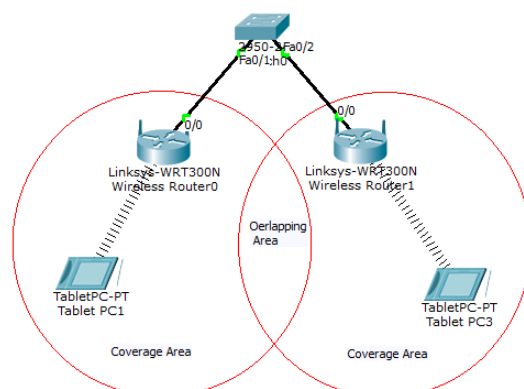
- ❖ **Wireless Adapter (Wireless LAN Adapter):** is a device that connects the client to the wireless medium.
- ❖ The purpose of the **access point** is to provide a bridge between the wireless LAN and wired network.
- ❖ The **access point:** is transmit/receiver unit (**transceiver**) that interconnects data from the wireless LAN to the wired network.
- ❖ An **IPSS - Independent Basic Service Set:** allows devices like laptops or mobile clients to connect directly to each other without needing an **Access Point**



- ❖ **BSS – Basic Service Set:** devices connect through a single Access Point to communicate or access wired resource.

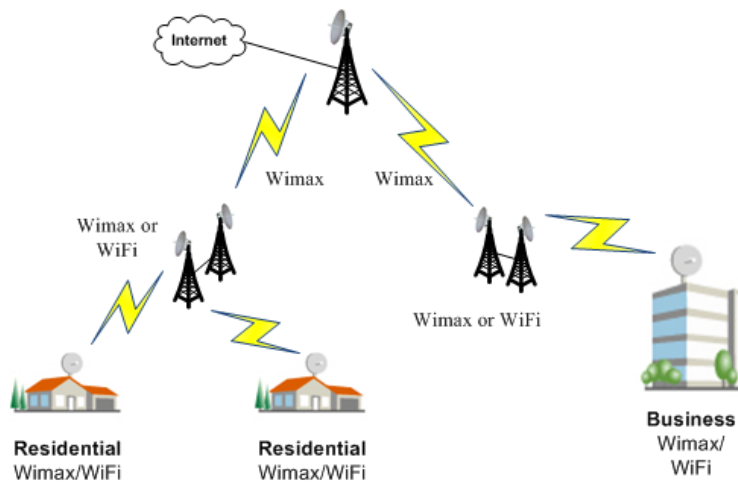


- ❖ **ESS – Extended Service Set:** connects two or more BSS through a WLAN controller (WLC) or RADIUS for broader network coverage.

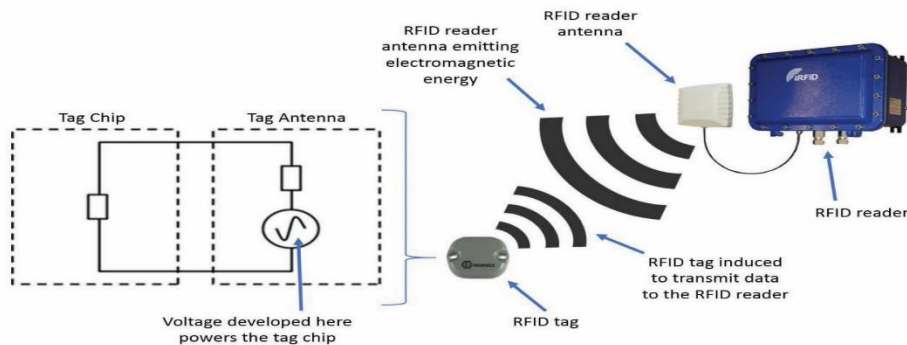


- ❖ **A wireless bridge:** is a popular choice for connecting LANs (similar network protocol) together even if the LANs are mile apart.
- ❖ the wireless LAN devices use an **SSID** to identify what wireless data traffic is allowed to connect to the network.
- ❖ **SSID – Service Set Identifier:** is the name of your wireless network, also known as **Network Identifier**.
- ❖ Outdoor site survey key points
 - ✚ Electrical power
 - ✚ Connection back to the home network
 - ✚ Antenna selection
 - ✚ Bandwidth supported
 - ✚ RF coverage
- ❖ Indoor Site survey key points
 - ✚ Electrical power
 - ✚ Wire network connection point(s)
 - ✚ Access point placement
 - ✚ RF coverage - user mobility
 - ✚ Bandwidth supported
- ❖ **A Site survey:** was conducted for determining the placement of an access point for wireless connectivity in a building.
 - ✚ The goal is to ensure mobile client access throughout the building
- ❖ A Piconet is an ad hoc network up to eight Bluetooth devices, with one device acting as the **master**, providing synchronization clock while others are **slaves**. e.g. Computers, keyboards, headsets.

- ❖ **WiMAX – Worldwide Interoperability for Microwave Access:** is a broadband wireless system and has been developed for use as **Broadband Wireless Access (BWA)** or fixed and mobile stations.



- ❖ **RFID** – is a technology that uses radio waves to track and identify people, animals, objects, and shipments.



- ❖ RFID consist of two things

- **Reader:** is a device that uses radio frequency waves to wirelessly transfer data between itself and RFID tag.
 - The **Reader** is also known as **Transceiver** which combines both transmitter and receiver along with antenna

- **Backscatter:** is when radio waves hit an RFID tag and bounce back to the transmitter with the tag's unique ID information.



- ✚ **RFID tag:** is also called **RF Transponder**, that have integrated antenna radio electronics.



- ❖ The three parameters that define RFID system are
 - ✚ Powering the tag
 - ✚ Frequency of operation
 - ✚ Communication protocol (air interface protocol)
- ❖ RFID are classified into three types based in powering tag
 - ✚ Passive
 - ✚ Semi-active
 - ✚ Active

- ❖ RFID typically use three frequency bands for operation
 - ✚ **Low frequency – LF**
 - ✚ **High Frequency - HF**
 - ✚ **Ultra-high Frequency - UHF**
- ❖ **Slotted Aloha:** is air interface protocol for RFID tags, similar to Ethernet.
- ❖ **CDMA – Code Division Multiple Access:** a communication system using spread-spectrum techniques to multiplex multiple signals within a single channel.
- ❖ **LTE – Long Term Evaluation:** a 4G wireless standard designed to provide 10 times faster speeds than 3G networks.

Chapter 4 Interconnecting the LANs

- ❖ Interconnecting LANs in a campus network or even interconnecting LANs in a WAN incorporate similar concepts and issues.
- ❖ The **Campus Network** is a collection of two or more interconnected LANs.
 - ✚ These Networks can be **inside building** or **across several buildings**.
- ❖ A **Hub** is a basic networking device that connects multiple computer.
 - ✚ It operates physical layer 1
 - ✚ It has no intelligence (dumb device)
 - ✚ It uses broadcasting
 - ✚ Shares bandwidth among connected devices
 - ✚ Working with 0s and 1s
- ❖ A bridge is used in computer network to interconnect two LANs together to separate network segments.

- + **A segment** is a section of a network separated by bridges, switches, and routers.
 - + Bridge is layer 2 device in the OS model meaning that it is used to use MAC address information to make a decision regarding forwarding data packets.
 - + Bridges use MAC address to create a bridging table that maps MAC address to specific ports.
- ❖ All networking devices (e.g., computers) contain **ARP – Address Resolution Protocol**: is a temporary storage of MAC address recently contacted. And it's also called **ARP table**.
- + The **ARP cache** stores MAC addresses of devices a computer has recently communicated with.

```
c:\arp -a
```

No ARP Entries Found

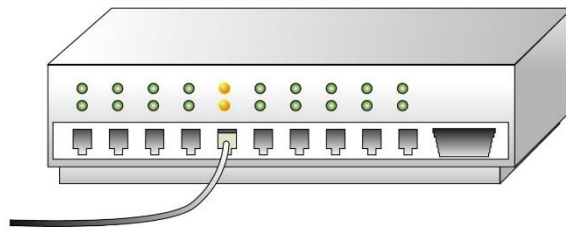
- + This message is generated if all of the ARP entries have expired or deleted.
- ❖ **Switches** are networking devices operating at layer 2 or a data link layer of the OSI model.
- + They connect devices in a network and use packet switching to send, receive, or forward between them.
 - + It uses MAC address to direct data and supports **unicast** (one-to-one), **multicast** (one-to-many), and **broadcast** (one-to-all) communication.

✚ There are two types of switch

- **Unmanaged switch:** simple, low-cost switches for home or small networks that work as soon as they're plugged in.
- **Managed switch:** more expensive and customizable switches used in larger networks, offering more control and features.

❖ A Networking connection can be verified by checking the **link light** on the hub.

✚ If it's the devices are connected, if it's off, there may be a cable or hardware issue.



❖ Benefits of Switches

- ✚ Less network congestion
- ✚ Faster data transfers
- ✚ Excellent manageability

❖ **Content-Addressable Memory (CAM):** special type of computer memory used in certain very high speed searching applications.

✚ It's used by switches to store MAC address and port mappings for fast lookups.

✚ Switches limit the amount of time address and port information are stored in CAM; this is called **Aging time**.

❖ The **Dynamic Address** tab shows MAC addresses that were automatically assigned when a device connected to a port.

❖ The **Static Address** tab shows the MAC address that were manually assigned and won't change.

- ❖ The **Secure** tab shows ports that are locked, if a device with a different MAC address tries to connect, the port will turn off automatically.



- ❖ Switching modes are

- **Store-and-forward:** the switch receives the entire data frame before deciding where to send it, ensuring the frame is complete and error-free

- Advantage and disadvantage of Store-and-forward
 - **Pros:** ensures error-free data
 - **Cons:** longer packets increase delay due to processing time.

- **Cut-through:** the switch start forwarding the data packet to its destination as soon as it reads the destination MAC address, reducing latency. There are two forms of cut-through

- **Fast-Forward:** Minimizes latency by sending the packet immediately after reading the destination MAC address
 - **Fragment-Free:** Filters out fragment collisions and discards them for better reliability.
 - **Fragment collision:** are collisions that occur within the first 64 bytes of the data packet.

❖ **Switch Latency:** is the length of time a data packet takes from the time it enters a switch until it exists.

❖ Steps to Assemble an office LAN

- **Document devices:** list all devices, including their MAC and IP addresses, and create simple network sketch
- **Connect devices:** use a star topology with twisted pair cables to link all devices.
- **Configure IP settings:** assign IP addresses to each computer based on the network administrator' plan

NOTE:

- NIC contains unique MAC and IP addresses
- MAC address are obtained by entering the ipconfig/all command from the command prompt

❖ The **Router** is layer 3 device in the OSI model, which means the router uses the network address to make routing decisions regarding forwarding the data packets.

✚ **Network address (Logical address):** is an IP address that identifies a network's location and the host within it, unlike physical address like MAC address.

✚ A router is configured to route data packets entering and leaving a LAN; rather than bridges use Ethernet to forward data.

✚ **Routers** are used to interconnect LANs in a campus network.

✚ It also make it possible to interconnect to LANs around the country and the world.

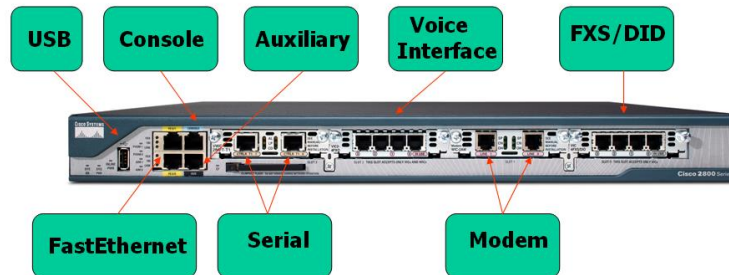
❖ Router inputs

✚ **Console input:** provides a serial (RS-232) or USB connection for initial router setup.

✚ **Auxiliary input:** allow a modem connection for remote access if the network is down using and RJ-45

✚ **Serial input**

- + Fast Ethernet input
- + Voice Interface Card
- + Modem
- + FXS/DID



❖ **A Gateway:** is a device that allows hosts in a LAN to connect to external networks.